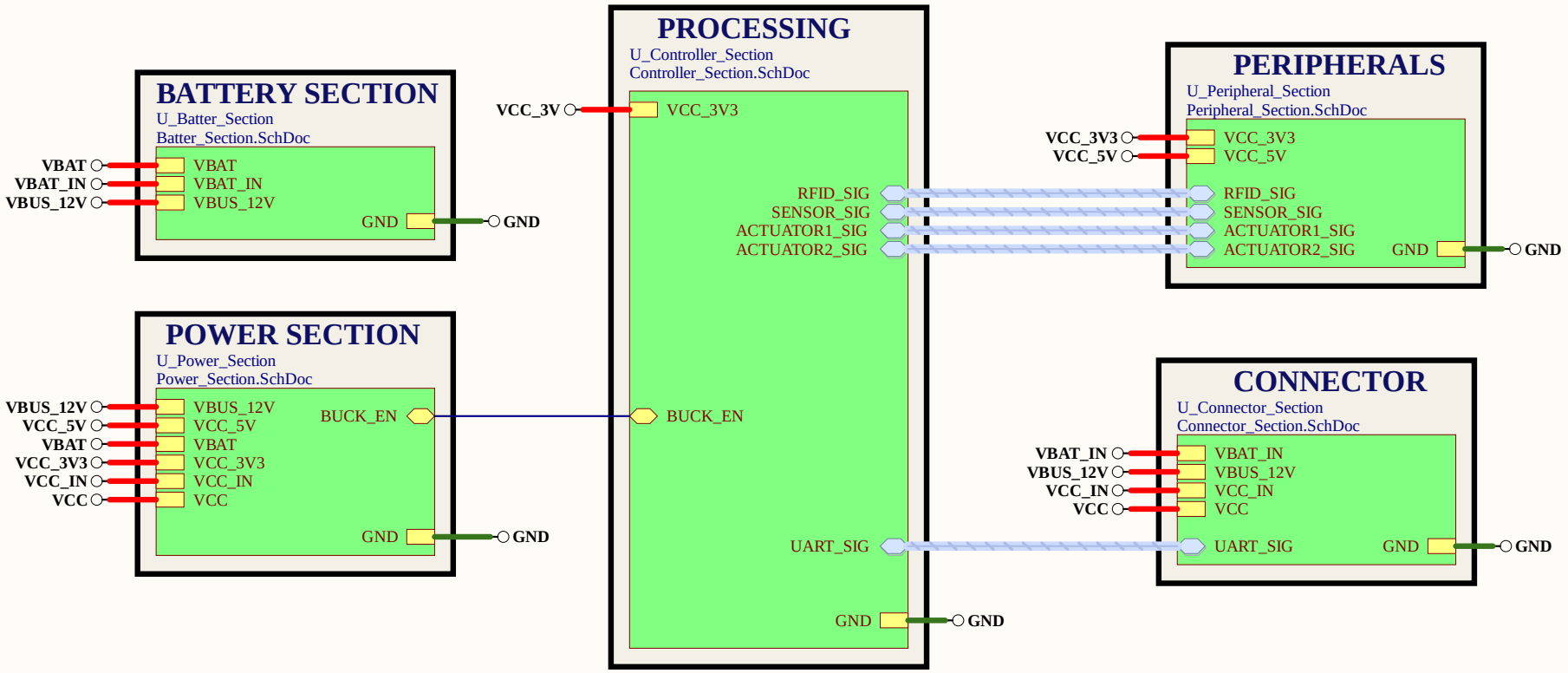
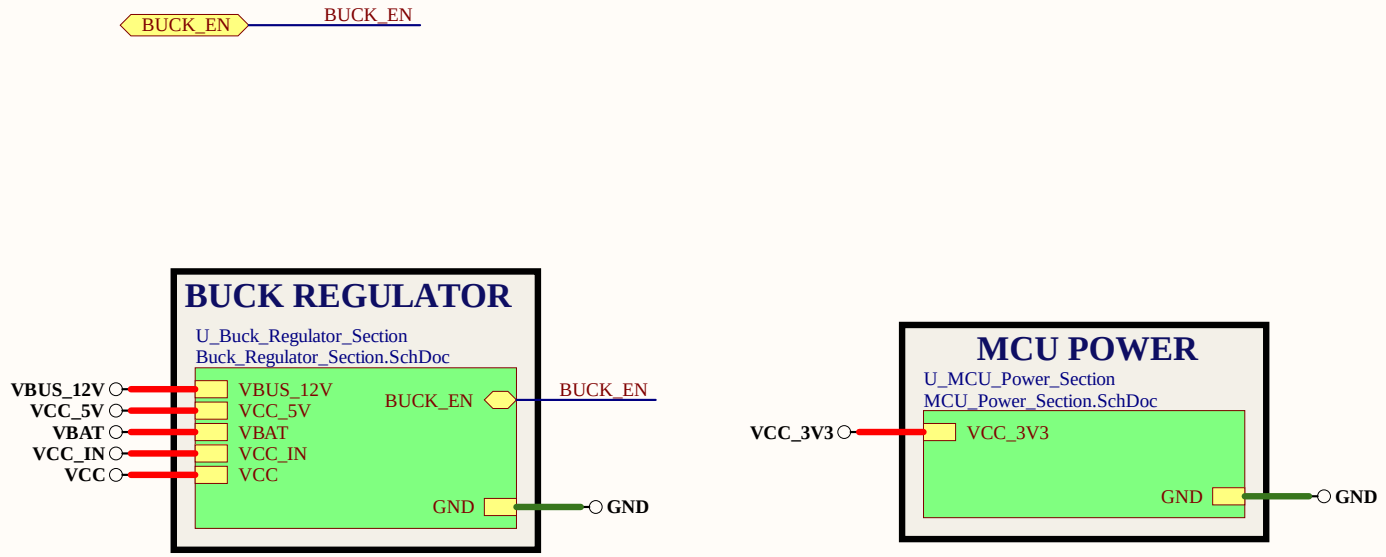
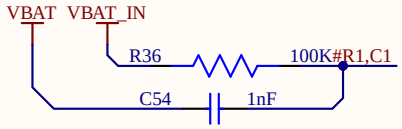
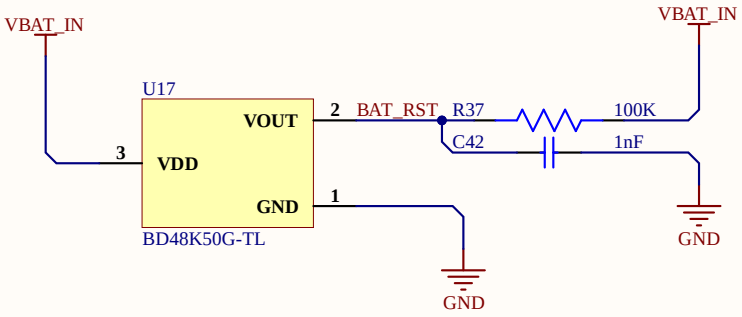

BATTERY SECTION
U_Batter_Section
Batter_Section.SchDoc
VBAT
VBAT_IN
VBUS_12V
GND
GND
POWER SECTION
U_Power_Section
Power_Section.SchDoc
VBUS_12V
VCC_5V
VBAT
VCC_3V3
VCC_IN
VCC
BUCK_EN
GND
GND
PROCESSING
U_Controller_Section
Controller_Section.SchDoc
VCC_3V
VCC_3V3
RFID_SIG
SENSOR_SIG
ACTUATOR1_SIG
ACTUATOR2_SIG
BUCK_EN
UART_SIG
GND
GND
PERIPHERALS
U_Peripheral_Section
Peripheral_Section.SchDoc
VCC_3V3
VCC_5V
RFID_SIG
SENSOR_SIG
ACTUATOR1_SIG
ACTUATOR2_SIG
GND
GND
CONNECTOR
U_Connector_Section
Connector_Section.SchDoc
VBAT_IN
VBUS_12V
VCC_IN
VCC
UART_SIG
GND
GND


BUCK_EN
BUCK_EN
BUCK REGULATOR
U_Buck_Regulator_Section
Buck_Regulator_Section.SchDoc
VBUS_12V
VCC_5V
VBAT
VCC_IN
VCC
VBUS_12V
VCC_5V
VBAT
VCC_IN
VCC
BUCK_EN
BUCK_EN
GND
GND
MCU POWER
U_MCU_Power_Section
MCU_Power_Section.SchDoc
VCC_3V3
VCC_3V3
GND
GND

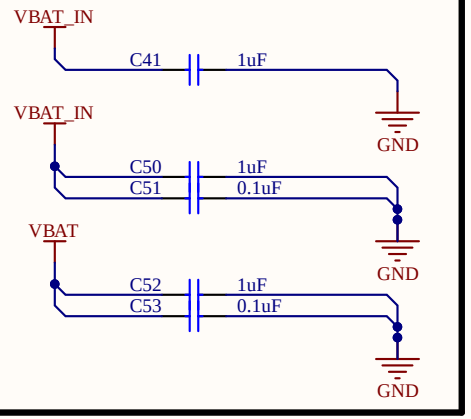
VOLTAGE DIVIDER



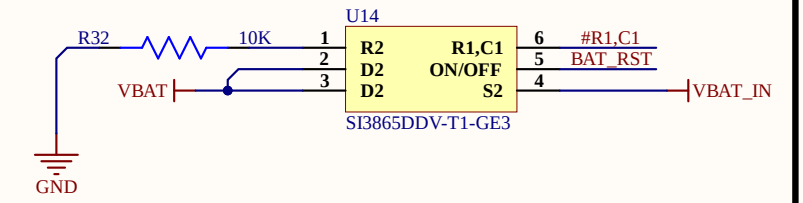
SUPERVISORY CIRCUIT



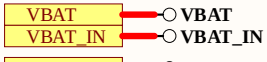
DECOUPLING CAPACITOR



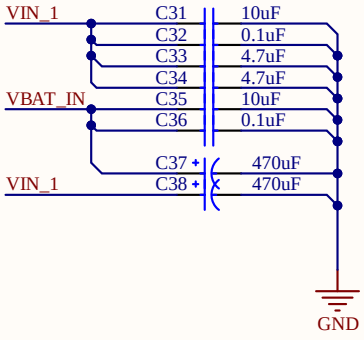
POWER SWITCH IC



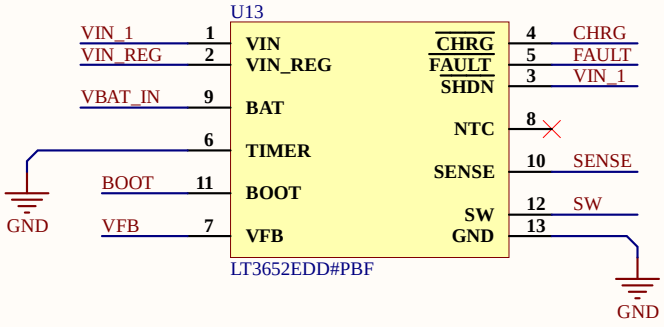
POWER NETS



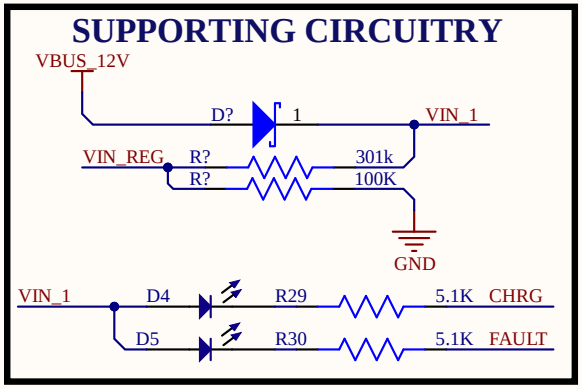
DECOUPLING CAPACITORS



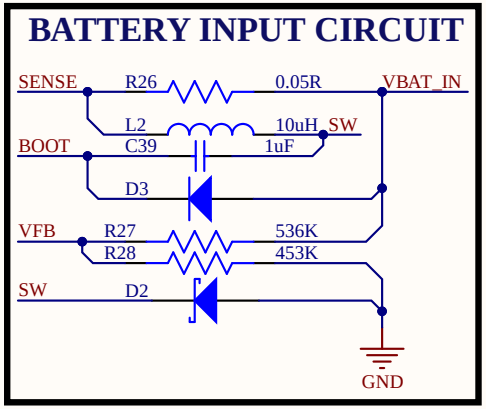
BATTERY CHARGING CIRCUIT



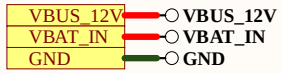
SUPPORTING CIRCUITRY

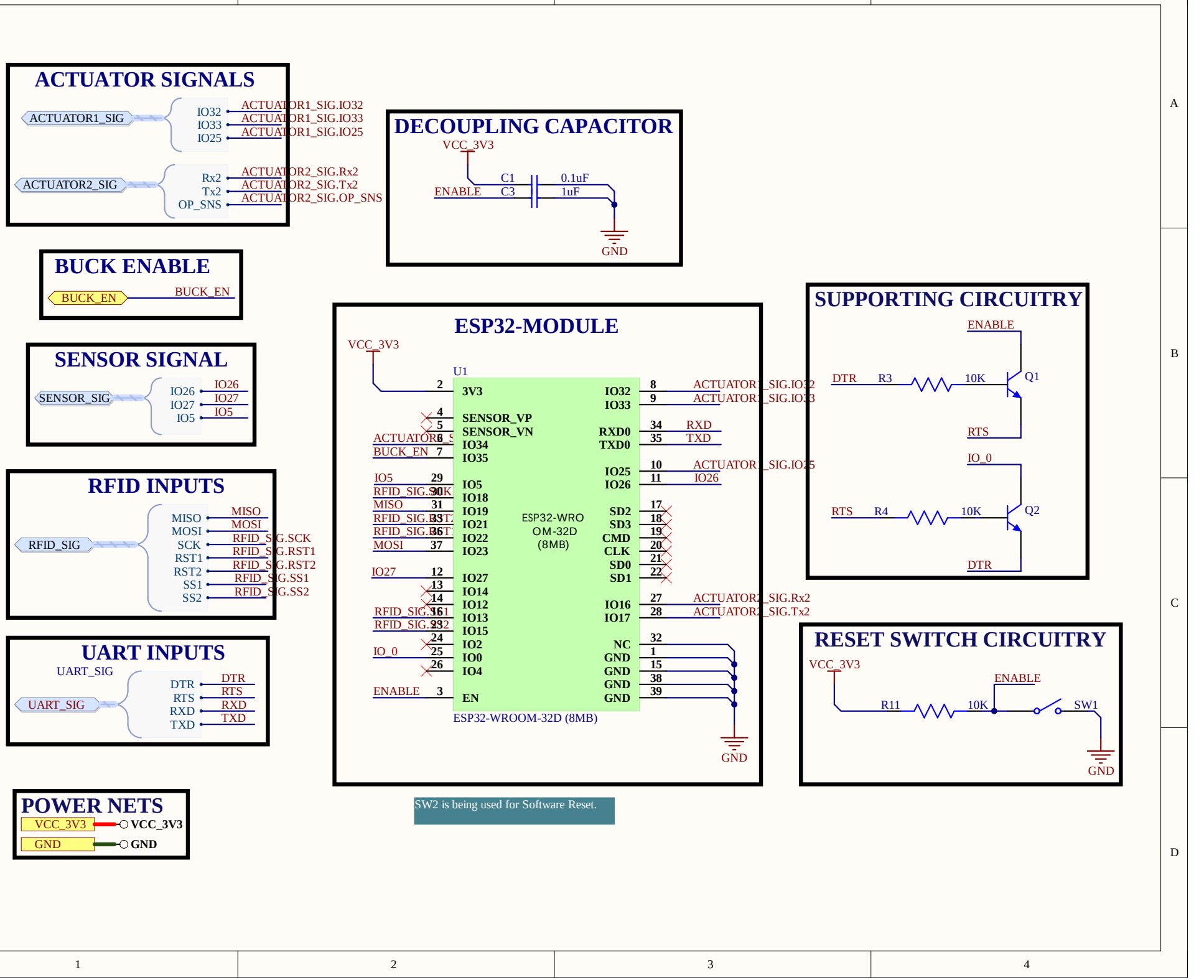


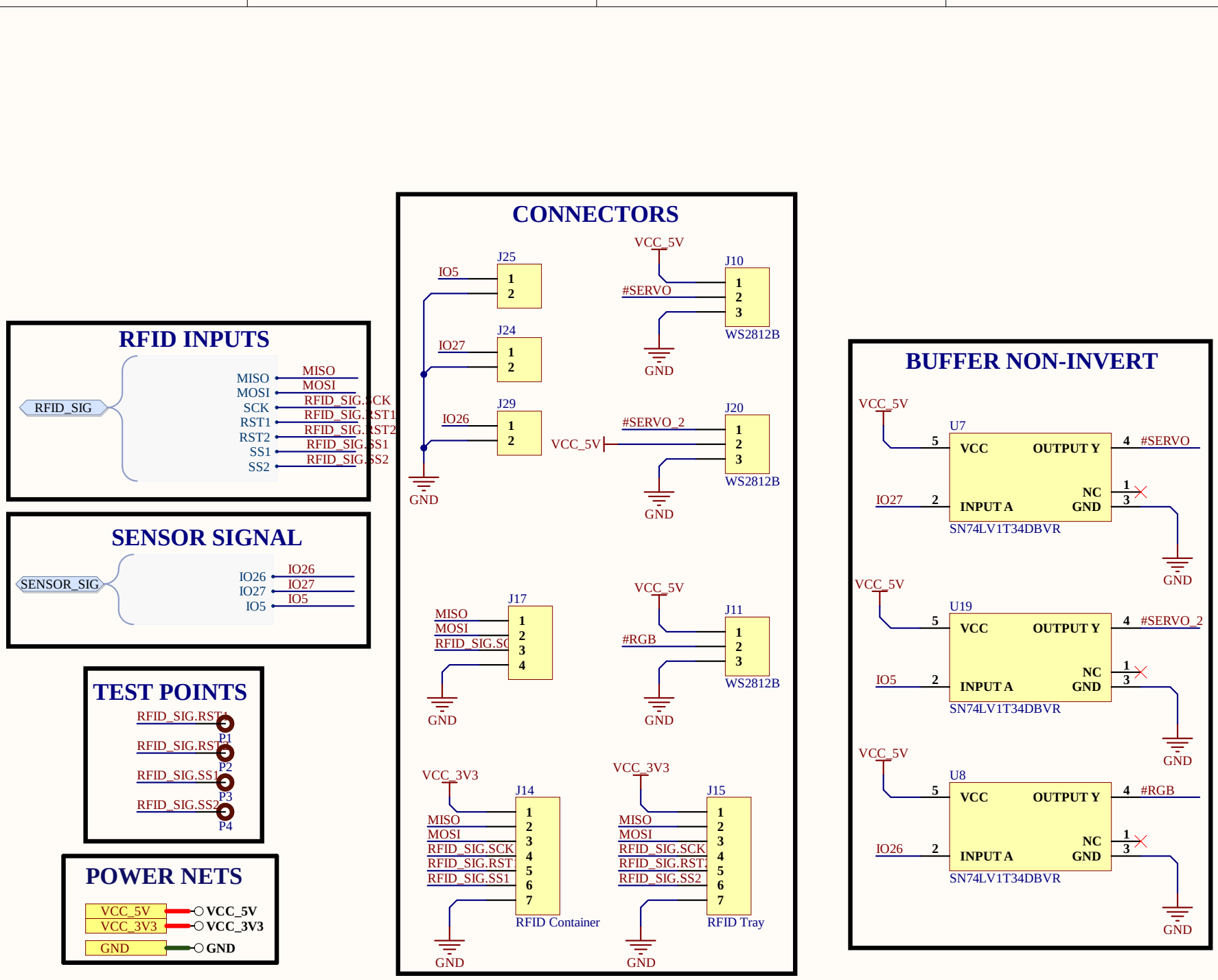
BATTERY INPUT CIRCUIT



POWER NETS



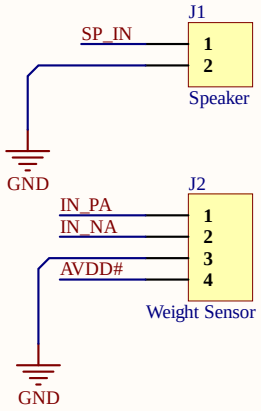

ACTUATOR SIGNALS
ACTUATOR1_SIG
IO32
IO33
IO25
ACTUATOR1_SIG.IO32
ACTUATOR1_SIG.IO33
ACTUATOR1_SIG.IO25
ACTUATOR2_SIG
Rx2
Tx2
OP_SNS
ACTUATOR2_SIG.Rx2
ACTUATOR2_SIG.Tx2
ACTUATOR2_SIG.OP_SNS
DECOUPLING CAPACITOR
VCC_3V3
C1
0.1uF
ENABLE
C3
1uF
GND
BUCK ENABLE
BUCK_EN
BUCK_EN
SENSOR SIGNAL
SENSOR_SIG
IO26
IO27
IO5
IO26
IO27
IO5
RFID INPUTS
RFID_SIG
MISO
MOSI
SCK
RST1
RST2
SS1
SS2
MISO
MOSI
RFID_SIG.SCK
RFID_SIG.RST1
RFID_SIG.RST2
RFID_SIG.SS1
RFID_SIG.SS2
UART INPUTS
UART_SIG
UART_SIG
DTR
RTS
RXD
TXD
DTR
RTS
RXD
TXD
POWER NETS
VCC_3V3
VCC_3V3
GND
GND
ESP32-MODULE
VCC_3V3
U1
2
3V3
4
5
SENSOR_VP
SENSOR_VN
6
IO34
BUCK_EN
7
IO35
IO5
29
IO5
30
IO18
MISO
31
IO19
33
IO21
RFID_SIG.RST2
36
IO22
MOSI
37
IO23
IO27
12
IO27
13
IO14
14
IO12
16
IO13
23
IO15
24
IO2
IO_0
25
IO0
26
IO4
ENABLE
3
EN
ESP32-WROOM-32D (8MB)
IO32
IO33
RXD0
TXD0
IO25
IO26
SD2
SD3
CMD
CLK
SD0
SD1
IO16
IO17
NC
GND
GND
GND
GND
8
9
34
35
10
11
17
18
19
20
21
22
27
28
32
1
15
38
39
ACTUATOR1_SIG.IO32
ACTUATOR1_SIG.IO33
RXD
TXD
ACTUATOR1_SIG.IO25
IO26
ACTUATOR2_SIG.Rx2
ACTUATOR2_SIG.Tx2
GND
SW2 is being used for Software Reset.
SUPPORTING CIRCUITRY
ENABLE
DTR
R3
10K
Q1
RTS
IO_0
RTS
R4
10K
Q2
DTR
RESET SWITCH CIRCUITRY
VCC_3V3
ENABLE
R11
10K
SW1
GND


RFID INPUTS
RFID_SIG
MISO
MOSI
SCK
RST1
RST2
SS1
SS2
SENSOR SIGNAL
SENSOR_SIG
IO26
IO27
IO5
TEST POINTS
RFID_SIG.RST1
P1
RFID_SIG.RST2
P2
RFID_SIG.SS1
P3
RFID_SIG.SS2
P4
POWER NETS
VCC_5V
VCC_3V3
GND
CONNECTORS
J25
J24
J29
J10
J20
J17
J11
J14
J15
WS2812B
RFID Container
RFID Tray
#SERVO
#SERVO_2
#RGB
BUFFER NON-INVERT
U7
U19
U8
VCC
OUTPUT Y
INPUT A
NC
SN74LV1T34DBVR

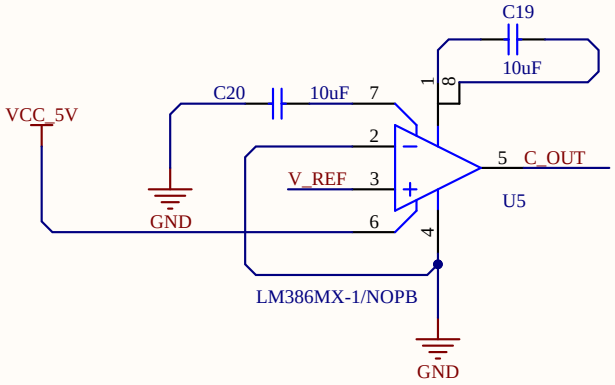
ACTUATOR1 SIGNAL



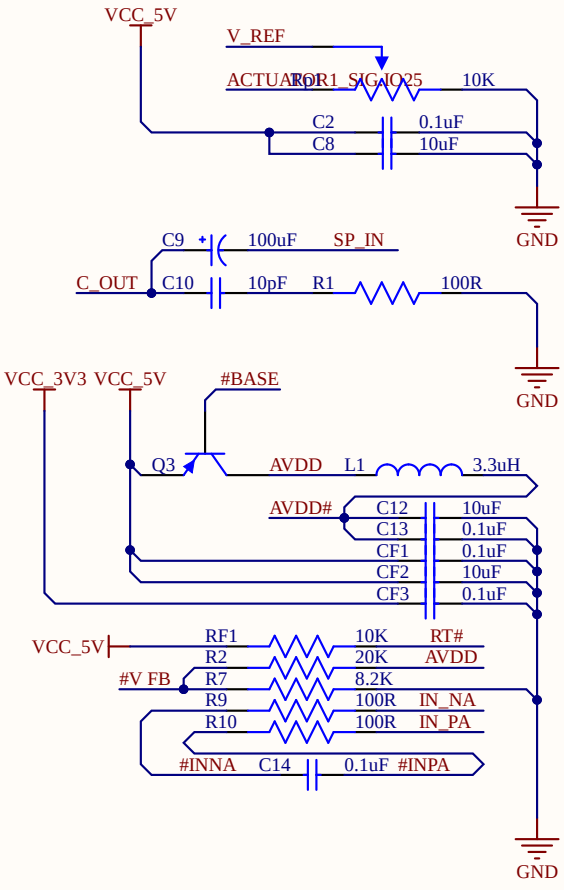
CONNECTORS



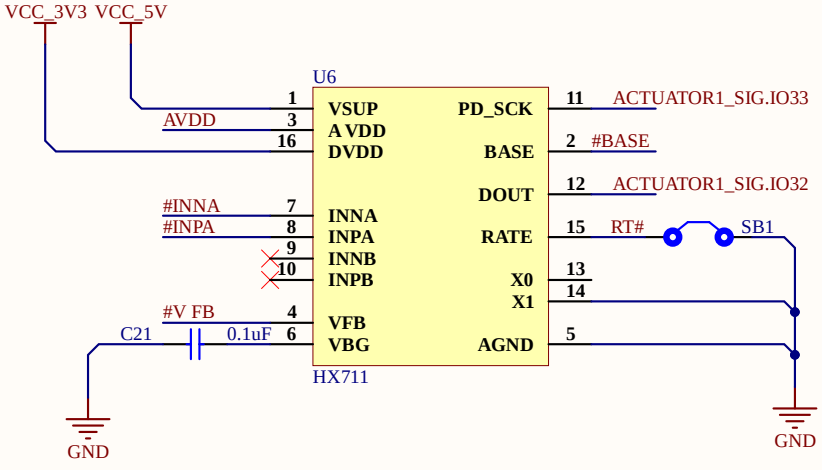
POWER CONVERTOR SIGNAL



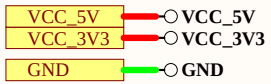
SUPPORTING CIRCUITRY

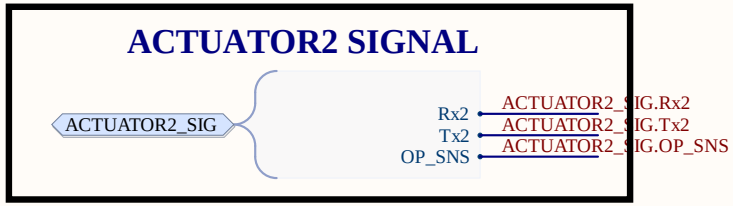


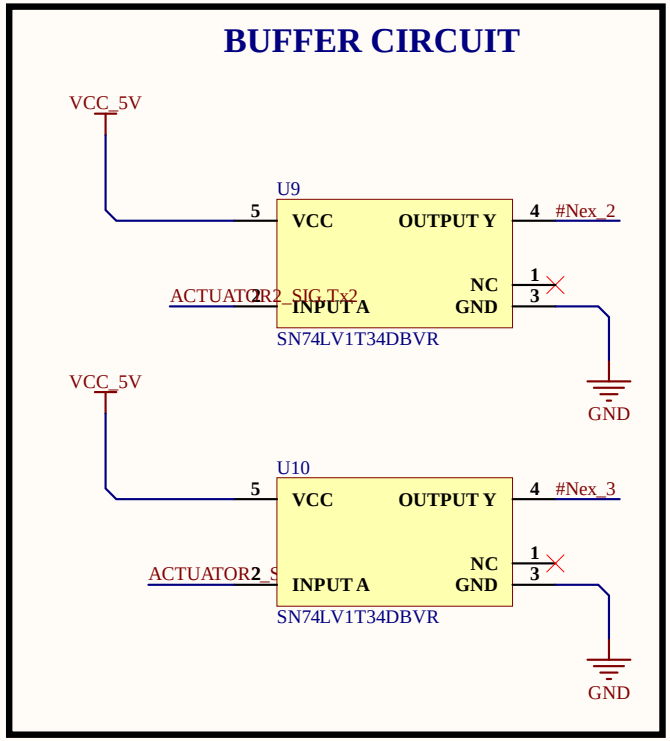
ACTUATOR1 SIGNAL

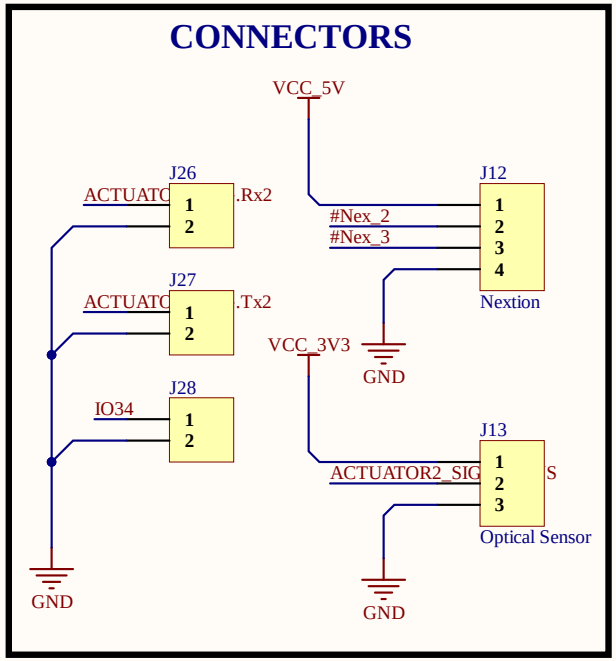


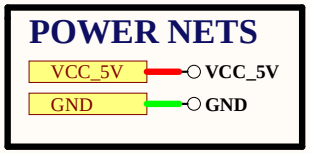
POWER NETS

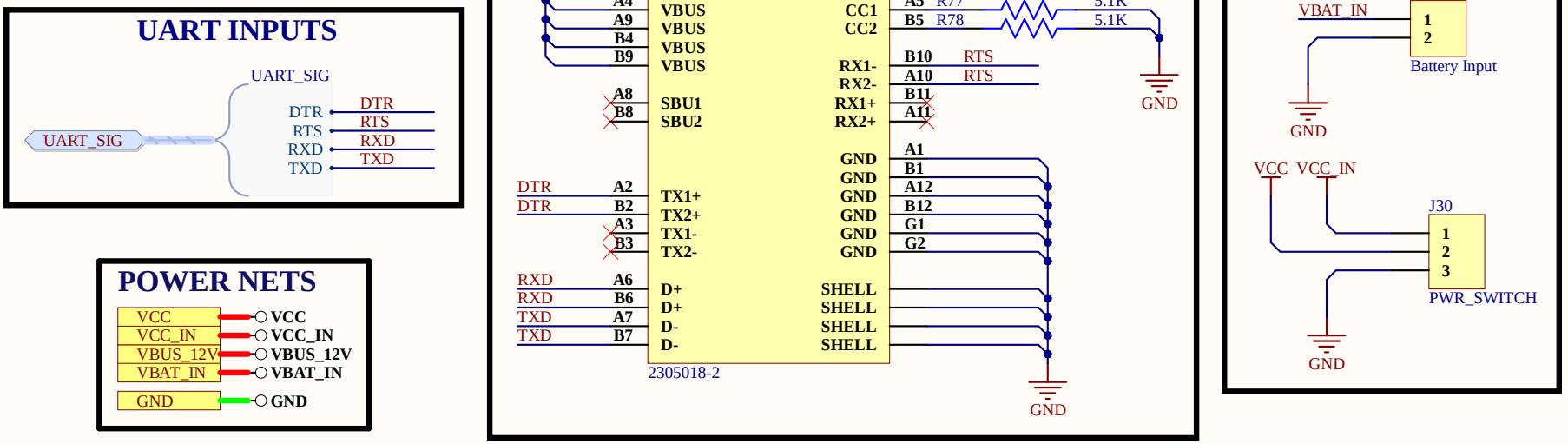


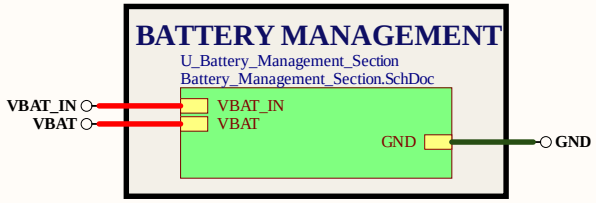

ACTUATOR2 SIGNAL
ACTUATOR2_SIG
Rx2
Tx2
OP_SNS
ACTUATOR2_SIG.Rx2
ACTUATOR2_SIG.Tx2
ACTUATOR2_SIG.OP_SNS

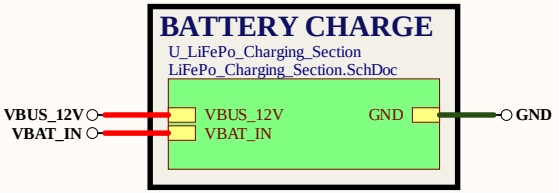

BUFFER CIRCUIT
VCC_5V
U9
5
VCC
OUTPUT Y
4
#Nex_2
NC
1
2
INPUT A
GND
3
SN74LV1T34DBVR
GND
VCC_5V
U10
5
VCC
OUTPUT Y
4
#Nex_3
NC
1
INPUT A
GND
3
SN74LV1T34DBVR
GND

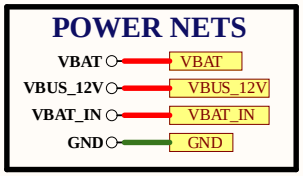

CONNECTORS
VCC_5V
J26
1
2
J27
1
2
J28
1
2
IO34
GND
J12
1
2
3
4
#Nex_2
#Nex_3
Nextion
GND
VCC_3V3
J13
1
2
3
ACTUATOR2_SIG
Optical Sensor
GND
GND


POWER NETS
VCC_5V
VCC_5V
GND
GND


UART INPUTS
UART_SIG
UART_SIG
DTR
RTS
RXD
TXD
DTR
RTS
RXD
TXD
POWER NETS
VCC
VCC_IN
VBUS_12V
VBAT_IN
GND
VCC
VCC_IN
VBUS_12V
VBAT_IN
GND
USB CONNECTOR
VBUS_12V
J16
A4
A9
B4
B9
VBUS
VBUS
VBUS
VBUS
A8
B8
SBU1
SBU2
DTR
DTR
A2
B2
A3
B3
TX1+
TX2+
TX1-
TX2-
RXD
RXD
TXD
TXD
A6
B6
A7
B7
D+
D+
D-
D-
2305018-2
CC1
CC2
RX1-
RX2-
RX1+
RX2+
GND
GND
GND
GND
GND
GND
SHELL
SHELL
SHELL
SHELL
A5
R77
5.1K
B5
R78
5.1K
B10
RTS
A10
RTS
B11
A11
GND
A1
B1
A12
B12
G1
G2
CONNECTORS
J3
VBAT_IN
1
2
Battery Input
GND
VCC
VCC_IN
J30
1
2
3
PWR_SWITCH
GND


BATTERY MANAGEMENT
U_Battery_Management_Section
Battery_Management_Section.SchDoc
VBAT_IN
VBAT
VBAT_IN
VBAT
GND
GND

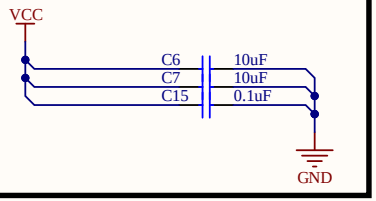

BATTERY CHARGE
U_LiFePo_Charging_Section
LiFePo_Charging_Section.SchDoc
VBUS_12V
VBAT_IN
VBUS_12V
VBAT_IN
GND
GND


POWER NETS
VBAT
VBUS_12V
VBAT_IN
GND
VBAT
VBUS_12V
VBAT_IN
GND

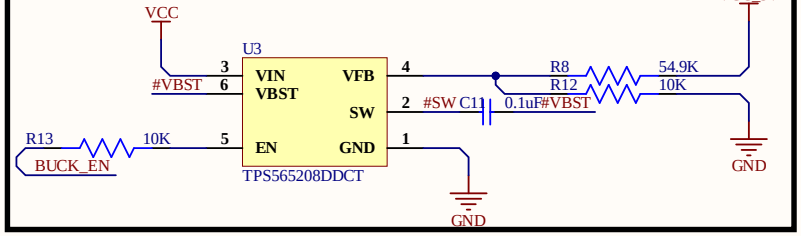
BUCK_EN

BUCK_EN

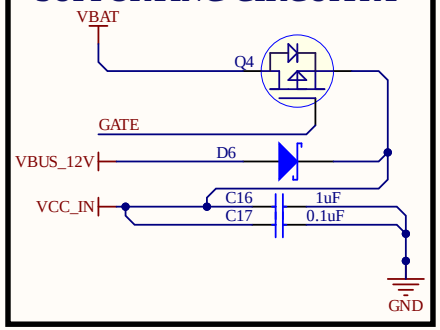
DECOUPLING CAPACITOR



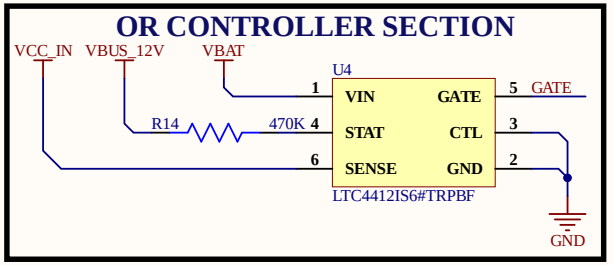
BUCK SWITCHING REGULATOR CIRCUIT



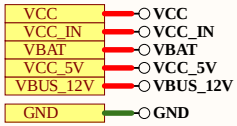
SUPPORTING CIRCUITRY



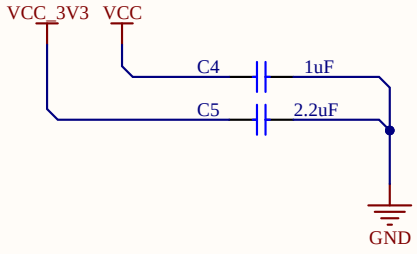
OR CONTROLLER SECTION



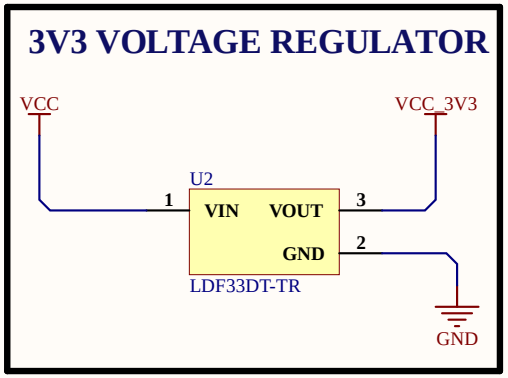
POWER NETS



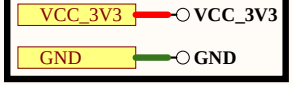
DECOUPLING CAPACITOR

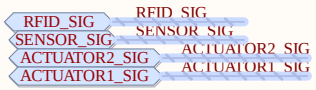


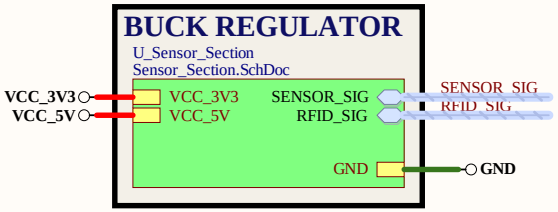
3V3 VOLTAGE REGULATOR

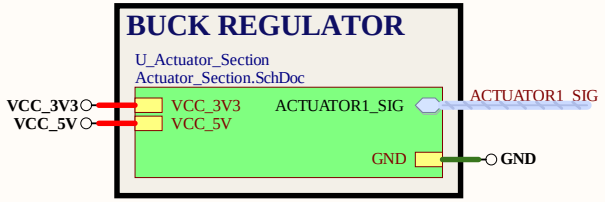


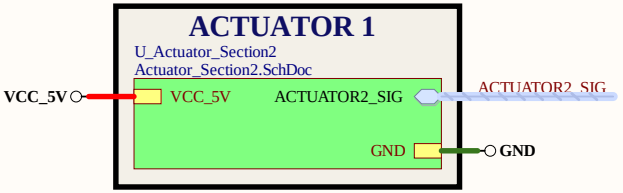
POWER NETS

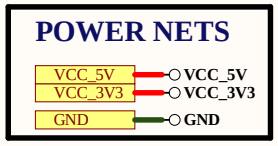



RFID_SIG
SENSOR_SIG
ACTUATOR2_SIG
ACTUATOR1_SIG


BUCK REGULATOR
U_Sensor_Section
Sensor_Section.SchDoc
VCC_3V3
VCC_5V
VCC_3V3
VCC_5V
SENSOR_SIG
RFID_SIG
SENSOR_SIG
RFID_SIG
GND
GND


BUCK REGULATOR
U_Actuator_Section
Actuator_Section.SchDoc
VCC_3V3
VCC_5V
VCC_3V3
VCC_5V
ACTUATOR1_SIG
ACTUATOR1_SIG
GND
GND


ACTUATOR 1
U_Actuator_Section2
Actuator_Section2.SchDoc
VCC_5V
VCC_5V
ACTUATOR2_SIG
ACTUATOR2_SIG
GND
GND


POWER NETS
VCC_5V
VCC_3V3
GND
VCC_5V
VCC_3V3
GND